

Summary of Findings – Titanic Dataset

1. Passenger Demographics

- **Age Distribution:** Most passengers were aged between **20–40 years**. Few were children or elderly. The distribution is slightly **right-skewed**.
 - **Fare Distribution:** Majority of fares were **under \$50**, with a few **high-paying outliers**, mostly in **1st class**.
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2. Survival Patterns

- **Survival & Age:** Younger passengers, especially **children**, had a **higher chance of survival**.
 - **Survival & Fare:** Passengers who paid **higher fares** were more likely to survive, indicating **class-based evacuation priorities**.
 - **Survival & Class:** **1st class passengers** had the **highest survival rate**, while **3rd class** had the lowest.
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3. Relationships Between Variables

- **Fare & Class:** Strong **negative correlation** — higher class meant higher fare.
 - **SibSp & Parch:** Positive correlation — those traveling with siblings often traveled with parents/children too.
 - **Class & Survival:** Passengers in **lower classes** had **significantly lower survival rates**.
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4. Key Visual Insights

- **Boxplots:** Survivors were generally **younger** and paid **higher fares**.
- **Scatterplots:** Showed clusters of survivors in **higher fare, younger age, and 1st class** groups.
- **Histograms:** Revealed **right-skewed distributions** for both **age** and **fare**.